Definitive Guide: Reworking the BeyondHulten Revision Strategy

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2026-08-19

1 Context and Scope

This document synthesises the results of Milestones A–E (completed 2026-08-18/19) and the newly implemented sticky-wage `:fixed` closure into a definitive guide for reworking the BeyondHulten manuscript. It is addressed to the revision strategy: what the data actually say, what narrative they support, and what must change in the paper.

It supersedes the planning assumptions in `WORKPLAN.md` and `WORKPLAN2.md` (both of which certified results that the broken model could not deliver) and operationalises the ROADMAP's Phase 6 go/no-go decisions with actual evidence.

Sources referenced: `ROADMAP.md`, `WORKPLAN3.md`, `PRELIMINARY-ASSESSMENT.md`, `README.md`, `roadmaps/vertdict.md`, `diag_fixed_closure.jl`, `diag_bridge_gauge.jl`, `diag_vd_honest.jl`.

1.1 Part I — What We Now Know (Verified Empirical Results)

1.1.1 1. The model is a proper equilibrium

Property	Result	Source
Sector-1 zero-profit	~1e-15 (restored)	Milestone A
All N zero-profit residuals	~1e-14…1e-9	Milestone A
Household budget exhaustion	exact (normalised demand)	Milestone A
Goods-market clearing	~1e-10	Milestone A
Solver retcode checked	fails loudly on MaxIters	Milestone A
Real GDP	consumption Tornqvist (B&F metric)	Milestone B

The original submission was not an equilibrium (household overspend 4.5%, missing zero-profit residual 10.9). That is now fixed. **This alone is a publishable methodological correction.**

1.1.2 2. The η (intersectoral mobility) channel is negligible

Bridge (aggregate GDP). Switching from immobile (η=0) to fully mobile (η=1) under a +30% supply shock gives:

Metric	Value
Gross-output Tornqvist	~0 pp (composition-dominated)
Consumption Tornqvist (B&F)	+0.005 to +0.034 pp
Interpretation	Second-order Harberger effect

Sobol variance share. η contributes **0.01%** of GDP variance across a full factorial grid over (η, ε, θ, σ).

Factor	S_f (first-order)	ST_f (total-order)
η	**0.0001** (0.01%)	0.0050
ε	0.0990 (9.9%)	0.1091

Factor	S_f (first-order)	ST_f (total-order)
θ	0.5186 (51.9%)	0.5239
σ	0.3720 (37.2%)	0.3746
1−ΣS_f (interactions)	0.0103 (1.0%)	

The old "88.4%" and "100%" results were artifacts of a non-equilibrium model with unnormalised demand shocks. **They are not recoverable.**

1.1.3 3. The binary wage regime (`:fixed` vs `:mobile`) matters materially

The sticky-wage (`:fixed`) closure pins the wage at 1.0 and lets employment absorb the shock. This is a different economic mechanism: **extensive margin (employment)**, not intensive margin (intersectoral reallocation).

Auton. demand mult	η=0 (mobile)	:fixed	**fix_bridge (pp)**	η_ratio
0.1	+0.002%	+0.251%	**+0.249**	**112×**
0.2	+0.009%	+0.635%	**+0.627**	**74×**
0.5	+0.047%	+1.203%	**+1.156**	—
1.0	+0.159%	+2.287%	**+2.129**	—
2.0	+0.467%	+4.228%	**+3.761**	—
5.0	+1.525%	+11.362%	**+9.837**	—
10.0	+3.146%	+21.661%	**+18.514**	—

The sticky-wage GDP response is **5–7×** larger than the full-employment baseline. Employment expands up to 22% above baseline (at mult=10.0). The η bridge, by contrast, never exceeds 0.01 pp.

Key finding: labour-market closure matters, but through the wage regime (flexible vs sticky) — not through the intersectoral mobility parameter (η).

1.1.4 4. Substitution elasticities (θ, σ) drive aggregate GDP

Within each wage regime, the aggregate response is determined by production structure. θ (intermediate substitution, 52%) and σ (consumption substitution, 37%) together account for 89% of GDP variance. This holds across all shock types tested (supply, demand-shift, autonomous, substitution and complementarity regimes).

1.2 Part II — What This Means for the Paper's Central Claims

1.2.1 The ROADMAP's hypothesis is half-right

> ROADMAP §1: "labour-market closure principally determines the feasible aggregate employment and real-GDP response; and substitution parameters principally affect sectoral allocation."

Claim	Supported?	Evidence
Labour closure determines aggregate GDP	**Partially** — through wage regime, not through η	fix_bridge 74–112× larger than η_bridge
Substitution parameters affect sectoral allocation	Not yet tested	Sobol on sectoral quantities not run
η (mobility) determines aggregate GDP	**No**	η S_f = 0.01%, bridge < 0.01 pp

The correct reformulation is:

> **"The aggregate GDP response is determined by the wage regime (flexible vs sticky) and by substitution elasticities (θ, σ). The intersectoral mobility parameter (η) is a second-order channel that only becomes material under extensions such as markups or complementarity."**

1.2.2 The three results the paper should now report

Result 1 — Methodological. The original model was not an equilibrium. A corrected, verified, budget-consistent version is now available. Residual diagnostics pass at machine precision. (This satisfies Rev 2's demand for a sound baseline.)

Result 2 — Negative but precise. The intersectoral mobility elasticity (η) is not a driver of the aggregate multiplier. Its variance share is <0.1% and the mobility bridge is <0.01 pp. The earlier reported "η dominance" (88.4%, 100%) was an artifact of model misspecification. The competitive one-factor CES model does not support the claim that labour mobility determines the output effects of a sector-specific demand programme.

Result 3 — Positive. The **binary wage regime** (flexible vs sticky) is material. A sticky wage amplifies the GDP response 5–7× over full employment because the shock is absorbed through employment expansion rather than wage adjustment. This identifies the correct labour-market margin for policymakers: the question is not "how mobile is labour?" but "is the economy at full employment or in an unemployment regime?"

1.3 Part III — Manuscript Restructuring

1.3.1 New proposed structure

Mapping from ROADMAP §7, updated for the new findings:

Section	Content	ROADMAP §
1. Introduction	Labour-market closure matters — but through the WRONG margin. The original η narrative was an artifact. State the corrected result.	§7.1
2. Related literature	IO–CGE relationship; labour closure taxonomy (mobility vs wage rigidity); B&F 2019/2022; CGE sensitivity analysis.	§7.2
3. Data and accounting	German 2019 IO table, calibration, Tornqvist real GDP, residual validation.	§7.3
4. Model and closures	Shared CES core. Two labour dimensions: (i) mobility (η continuum), (ii) wage regime (`:mobile`/`:fixed`). Financing closure.	§7.4
5. The mobility channel (η) is negligible	Bridge < 0.01 pp. Sobol S_f = 0.01%. The original "88.4%" was an artifact. Show the corrected variance decomposition.	§7.5
6. The wage regime channel is material	`:fixed` vs `:mobile`: 5–7× GDP amplification. Employment response. Policy interpretation.	§7.6
7. What drives aggregate GDP: θ and σ	Sobol results show 89% of GDP variance from substitution elasticities. Distinguish aggregate (elasticity-driven) from sectoral (not yet quantified).	§7.6
8. Robustness and limitations	Financing closures, shock magnitude, static horizon, one-factor simplification.	§7.8

Section	Content	ROADMAP §
9. Conclusion	Practical guidance: labour-market analysis should focus on the wage regime, not on the mobility elasticity. The IO multiplier is attainable under sticky wages; it is not under full employment.	§7.9

1.3.2 Headline results for the abstract

1. A corrected equilibrium 71-sector CGE model of the German housing transformation.
2. Intersectoral labour mobility (η) contributes **<0.1%** of GDP variance and a mobility bridge of **<0.01 pp** — a second-order Harberger effect.
3. The **binary wage regime** (sticky vs flexible) is a first-order channel: sticky wages amplify GDP **5–7×** through employment expansion.
4. Substitution elasticities (θ, σ) explain **89%** of GDP variance; the earlier "η = 88.4%" claim was an artifact.

1.3.3 What to remove from the manuscript

- All claims about the "η-bridge" as a first-order mechanism
- The "88.4%" or "η dominates" statements
- The "GO" certification language
- The idea that the IO multiplier requires η□∞ (it requires sticky wages)
- Any Type-I/Type-II multiplier framing that conflates mobility with wage rigidity
- Price-invariance claims (they were artifacts of the broken model)

1.3.4 What to add to the manuscript

- A clear distinction between **two labour-market margins**: intersectoral mobility (η) and wage regime (`:fixed` vs `:mobile`)
- The corrected variance decomposition (Sobol S_f, ST_f, no renormalisation)
- The `:fixed` closure results across shock sizes (table from §3 above)
- A note that the original results were from a non-equilibrium model, now corrected
- A statement that the paper does **not** claim to discover the IO–CGE bridge (per ROADMAP §2)

1.4 Part IV — ROADMAP Compliance Check

1.4.1 Phase 6 go/no-go decisions (now answerable)

Question	Answer	Implication
Does labour closure affect aggregate outcomes materially?	**Yes — through wage regime; No — through η**	Abandon "η closure dominance"; report wage-regime dominance
Do CES elasticities affect sectoral incidence?	Not yet tested	Run sectoral Sobol (Phase 7 below)
Does the one-factor model yield credible policy interpretation?	**As a benchmark, yes**	Present as methodological, not policy-evaluative
Does the IO endpoint emerge under limiting assumptions?	Not tested	Requires `:fixed` closure + Leontief coefficients

1.4.2 Outstanding ROADMAP requirements

Requirement	Status	Action
§4.1 Accounting consistency	Not yet reconciled	Phase 1 (new work)

Requirement	Status	Action
§4.3 Financing closure documented	Partially	Autonomous demand implemented, but not as a distinct tax/borrowing
§4.4 Labour closure taxonomy	Partially	`:mobile` and `:fixed` exist; `:unemployment` complementarity not implemented
§5 Closure D (IO endpoint)	Not implemented	Requires `:fixed` + CES→Leontief limit
§6 Phase 1 (account reconciliation)	Not started	Separate workstream
§6 Phase 2 (policy experiment)	Not started	Separate workstream

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1.5 Part V — Remaining Work (Next Steps for the Revision)

1.5.1 Short-term (can be done in this environment)

1. **Record the `:fixed` closure results to CSV**
   - Save `output/wage_regime_comparison.csv` with the mult-sweep table
2. **Update WORKPLAN3.md** — add the `:fixed` closure as a sub-milestone
3. **Run Sobol on sectoral quantities** — confirm whether η matters for sectoral allocation even if it doesn't for aggregate GDP

1.5.2 Medium-term (implementation work, 1–2 weeks)

4. **Phase 1 — Accounting reconciliation** (ROADMAP §6, Phase 1)
   - Benchmark accounting report from the German IO table
   - Separate domestic vs imported intermediate/final uses
   - Treatment of taxes and subsidies
5. **Phase 2 — Policy experiment** (ROADMAP §6, Phase 2)
   - Fix the sectoral investment vector
   - Choose and implement the principal financing closure
   - Distinguish the four admissible experiment types
6. **Milestone C — Solver stability** (Cobb-Douglas limit)
   - Guard against DomainError at $\varepsilon \to 1$ with sign-safe real powers

1.5.3 Longer-term (manuscript work, 2–3 weeks)

7. **Rewrite the manuscript** per Part III above
8. **Generate new tables and figures** from the corrected model
9. **Write the response-to-reviewers letter** — explain how the earlier results were artifacts and what the corrected model shows
10. **Run the :fixed closure with η=1** (to confirm the :fixed results are not η-driven within the sticky-wage regime)

1.6 Part VI — Summary: The Corrected Story

The original paper claimed:

- "Labour supply elasticity η drives the GDP response (+1.5% under $\eta \to \infty$)"
- "η accounts for 88.4% of output variance"
- "The model is a valid equilibrium"

All three were false. The corrected model shows:

- **η (mobility) drives <0.1%** of GDP variance; the bridge is <0.01 pp

- **The wage regime (sticky vs flexible) is the relevant labour-market margin**, amplifying GDP by 5–7× through employment
- **Substitution elasticities (θ, σ) explain 89%** of GDP variance
- The model is now a **verified equilibrium** with machine-precision residuals

This is a publishable result, but it requires the manuscript to abandon the original "η-dominance" narrative and replace it with a **three-part story**: (i) a corrected methodological baseline, (ii) a precise negative result about the mobility channel, and (iii) a positive result about the wage-regime channel and the dominance of substitution elasticities.